Take-Home Assessment - (Intern Developer – Proxy Infrastructure)

🎯 Goal

Demonstrate your ability to **learn independently, problem-solve, and implement a working system** related to proxy infrastructure.

📝 Task

Build a simple SOCKS5 proxy server in **Go** or **Node.js** that can:

1. Accept incoming client connections.
2. Forward traffic to the requested destination (basic tunneling).
3. Log each connection (source IP, destination host/port).
4. Allow configuration of:
   - Listening port (via env var or config file).
   - A hardcoded **username/password** for authentication.

📚 Constraints & Guidance

- You don't need to implement the **entire SOCKS5 RFC** — just enough for a client to connect, authenticate, and tunnel TCP traffic.
- Use only **standard libraries** if possible (e.g., `net` in Go, `net` in Node).

- You are free to **look up documentation, blog posts, or RFCs** — the key is how quickly you can **self-learn and apply**.

✅ Deliverables

1. A **GitHub repo** with:
    - Source code.
    - Minimal README with:
        - How to run the proxy.
        - Example test (e.g., using `curl` through the proxy to fetch `https://ipinfo.io`).
    - A short **reflection note** (2–3 paragraphs):
        - What you had to learn.
        - How you approached debugging.
        - What you would improve if given more time.

2. Video recording of your product demonstration

⏱ Time Expectation

This task is scoped to **6–8 hours of effort**, but you may take up to **3 days** to submit.

📊 Evaluation Criteria

- **Independence**: Did you figure things out without needing handholding?

- **Functionality**: Does the proxy actually forward connections and support authentication?
- **Code quality**: Is the code reasonably clean, modular, and documented?
- **Problem-solving mindset**: How well did you explain what you learned and overcame?
- **Resourcefulness**: Did you use docs, examples, or RFC references effectively?

📨 Assignment Submission

- Create a **Public GitHub Repository**
- Share the repository link using the form below
- Share a YouTube or Google Drive link with your video demo

Assignment Submission Form: https://airtable.com/app9kXuirRbKEJ88P/pagdXTgo0S7U3u4R9/form

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